



Interactive AI Systems

Meta's AI Chatbots

Meta's 2024 chatbots featured multimodal interactions, combining text, voice, and image inputs. Users could engage in conversational queries while sharing visual content for analysis. For example, uploading a photo of a product during a chat would prompt the AI to provide reviews, pricing, and availability.

These chatbots also supported personalized responses. By analyzing user preferences and behavior, the AI delivered contextually relevant recommendations, enhancing user satisfaction.

Smart Home Assistants

Google Assistant and Amazon Alexa deepened their multimodal integrations with smart home devices. In 2024, these assistants provided feedback through voice prompts, on-screen visuals, and haptic vibrations. For example, Alexa's smart displays offered step-by-step cooking instructions, combining verbal guidance with visual timers and video demonstrations.

This multimodal feedback improved accessibility for users with diverse needs. Haptic cues on smart displays aided users with hearing impairments, while detailed visuals supported those unfamiliar with verbal instructions.

Interactive Learning with AI Tutors

AI-powered learning platforms like Khan Academy introduced tools for multimodal education. These tools have proven especially effective in subjects like mathematics, science, and language arts at both secondary and higher education levels. Students could interact with AI tutors through spoken queries, written notes, and visual aids such as graphs and diagrams. For instance, a student struggling with algebra could receive step-by-step explanations in text, accompanied by animated visualizations.

These systems also adapted to individual learning styles, offering tailored feedback and resources. In real-world applications, they have been used to address gaps in remote learning by

providing personalized attention and supplementary resources to students in underprivileged areas. By combining modalities, they provided a more engaging and effective educational experience. AI-powered learning platforms like Khan Academy introduced tools for multimodal education. Students could interact with AI tutors through spoken queries, written notes, and visual aids such as graphs and diagrams. For instance, a student struggling with algebra could receive step-by-step explanations in text, accompanied by animated visualizations.

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AI-Powered Design and Architecture

SpaceMaker AI brought innovation to architecture by supporting multimodal design inputs. Architects could describe concepts verbally, sketch ideas, and upload reference images. The AI synthesized these inputs to generate comprehensive building designs, factoring in environmental considerations like light, noise, and energy efficiency.



This approach streamlined the iterative design process, enabling architects to explore multiple configurations quickly. It also supported collaboration by visualizing designs in real time, facilitating client feedback.

Interactive Storytelling

Narrative-driven platforms like ChatGPT Stories offered collaborative storytelling experiences. Users could provide prompts in text, voice, or images, and the AI generated subsequent story elements. For instance, a user could upload a landscape photo and request a plot set in that environment, with the AI crafting characters and events accordingly.

This multimodal interaction fostered creativity, allowing users to explore diverse narrative possibilities. It also catered to educators and game developers seeking dynamic, interactive content. 